

MAT 017B Calculus for Biology and Medicine

Course Information:

2018 Summer Session 2

Olson Hall 147

MTWR 4:10 - 5:50 PM

<https://www.math.ucdavis.edu/~mgaerlan/ss2.2018.017b.html>

Instructor:

Mikhail Gaerlan

mgaerlan@math.ucdavis.edu

MSB 2117 MTWR 3:10 - 4:00 PM



Course Information:

Course Description:

Introduction to integral calculus, elementary differential equations, and linear algebra via applications to biology and medicine. Fundamental theorem of calculus, techniques of integration including integral tables and numerical methods, improper integrals, elementary first order differential equations, linear systems, matrices, eigenvalues and eigenvectors, analytic geometry, applications in biology and medicine.

Textbook: C. Neuhauser, *Calculus for Biology and Medicine*, 3rd Edition, Pearson Custom Publ. ISBN 978-0-321-64468-8. This book is available online through [Inclusive Access](#).

Course Website: <https://www.math.ucdavis.edu/~mgaerlan/ss2.2018.017b.html>

All course material will be uploaded to the website. Only grades and announcements will be posted to [Canvas](#). The QR code above links to the course website.

Grades:

Grades will be posted on [Canvas](#) and [Gradescope](#). Regrade requests will be submitted through [Gradescope](#). Please accept the Gradescope course invitation sent to your UC Davis e-mail.

Outline:

Homework	5%
Quizzes	10%
Worksheets	15%
Midterms	40%
Final	30%

Homework: Homework will be graded based on effort and is to be submitted no later than the end of the class period on the due date. Homework questions will come from the book. No late homework will be accepted. Please write clearly, show your work, and box your answers. One homework will be dropped.

Quizzes: Quizzes will be administered during the first 15 minutes of class on the days they are scheduled. Quizzes will consist of 2-3 questions based on the assigned homework due the previous day along with the readings from the textbook. One quiz will be dropped.

Worksheets: Every Wednesday will be devoted to a worksheet to explore various computer-based applications of calculus. You will be assigned a new group every week, but each person must submit a separate worksheet due the following week. The worksheets will rely heavily on [WolframAlpha](#)[®].

Midterms and Final: No outside material or help may be used on any exam. A formula sheet will be provided. Midterm 1 will test material covered until August 14, while midterm 2 will test material covered from August 15 to August 28. The final exam will be cumulative.

Class Policies and Guidelines:

Calculator and Computers: No calculators or computers will be allowed during exams or quizzes. You may use a calculator for the homework but remember to show your work to get credit. You will need a laptop, tablet, or smartphone to access the [WolframAlpha®](#) website to fill out the worksheets on Wednesdays. No special software is required.

Academic Conduct: All students must adhere to the [Code of Academic Conduct](#). You are encouraged to work together on homework but be mindful that copied homework will look obvious to the reader. You will also be expected to work together on worksheets but each student must submit their own original work. In both cases, please submit the names of the classmates that you worked with. If you have any questions about what is allowed, feel free to ask me any questions. You may also contact the [Office of Student Support & Judicial Affairs](#).

Accommodations: If you have any unavoidable circumstances (e.g. family emergency, illness) that may cause you to miss an exam or quiz, please contact me as soon as possible to handle the situation. Students with disabilities should meet with me in person to confirm your arrangements. Contact the [Student Disability Center](#) for further questions.

Important Dates:

Last day to add classes	August 10
Last day to drop classes with refund	August 13
Midterm 1	August 16
Midterm 2	August 30
Final exam	September 13

Resources:

- [Help with Calculus](#)
<https://www.math.ucdavis.edu/resources/learning/calculus-help>
- [Student Academic Success Center](#)
<http://success.ucdavis.edu/services/mathematics.html>
- [CALCULUS.ORG](#)
<http://www.calculus.org>
- [The Calculus Page](#)
<http://www.ecalculus.org>

Syllabus

August 6	August 7	August 8	August 9
Antiderivatives and Integration	Fundamental Theorem of Calculus and Substitution Rule	Numerical Integration	Integration by Parts
Section 5.8, 6.1	Section 6.2, 7.1	Section 7.5	Section 7.2
August 13	August 14	August 15	August 16
Partial Fraction	Improper Integrals	Arc Length and Surfaces of Revolution	Midterm 1
Section 7.3	Section 7.4	Section 6.3	
August 20	August 21	August 22	August 23
Averages and Cumulative Change	Taylor Polynomials	Taylor Polynomials	First Order Differential Equations
Section 6.3	Section 7.6	Section 7.6	Section 8.1
August 27	August 28	August 29	August 30
Integrating Factors	Equilibria and Stability	Numerical Solutions of Differential Equations	Midterm 2
	Section 8.2	Section 7.6	
September 3	September 4	September 5	September 6
Solving Linear Systems	Matrices	Linear System Numerical Methods	Eigenvalues and Eigenvectors
Section 9.1	Section 9.2	Section 9.2	Section 9.3
September 10	September 11	September 12	September 13
Eigenvalues and Eigenvectors	Analytic Geometry	Review	Final Exam
Section 9.3	Section 9.4		